

LATENT PLANNING VIA EMBEDDING ARITHMETIC: A CONTRASTIVE APPROACH TO STRATEGIC REASONING

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ABSTRACT

Planning in high-dimensional decision spaces becomes more tractable when candidate decisions can be organized by learned structure rather than enumerated as raw states. This paper studies that principle in a setting where reliable state evaluations are available: chess positions labeled by Stockfish-derived win probabilities. We introduce SOLIS, a supervised-contrastive planning method that trains a transformer encoder to place positions with similar evaluations near one another and then uses a global *advantage vector* to orient the resulting space from losing to winning regions. At inference, SOLIS does not train a separate policy head, value head, or latent dynamics model; instead, it embeds legal successor positions and ranks them by projection along this evaluation-aligned direction. In chess, this representation-guided procedure attains competitive strength under constrained search, including 2500+ Elo with 5-ply search under 50 ms/move blitz conditions, and remains competitive in fixed-depth comparisons while expanding substantially fewer explicit nodes. These results support a focused claim: when evaluations can be distilled into representation geometry, shallow search over legal successors can use embedding arithmetic as a compact interface for strategic action selection.

1 INTRODUCTION

Planning in high-dimensional state spaces remains a central problem in artificial intelligence because the number of reachable continuations grows rapidly with each decision. Classical search and reinforcement-learning systems address this combinatorial pressure through explicit lookahead, learned policies, or value estimation, but these mechanisms can become computationally demanding when the action space is large or when effective play depends on long-horizon discrimination (Jiang et al., 2023; Gieselmann and Pokorný, 2022). Latent-planning methods offer a complementary strategy: they reason over compact learned representations rather than raw states, thereby attempting to preserve decision-relevant structure while reducing the effective complexity of action selection (Hafner et al., 2019; 2020; Schrittwieser et al., 2020).

We investigate whether an embedding space can serve not merely as an input to a value or policy module, but as the planning interface itself. We introduce **SOLIS** (*Strategic Optimization via Latent Interpolative Spaces*), a chess engine whose transformer encoder is trained with supervised contrastive learning to map Stockfish-evaluated positions into an evaluation-aligned latent space. In this representation, proximity reflects similarity of outcome, and displacement along a single global direction provides a heuristic ordering from disadvantage toward advantage. SOLIS operationalizes that geometry by ranking legal candidate moves according to their alignment with an “advantage vector” connecting losing and winning regions of the embedding space.

The resulting engine is empirically strong within the constrained evaluation regime studied here. With 5-ply search under 50 ms/move blitz conditions, SOLIS achieves an estimated Elo above 2500 against rating-capped Stockfish configurations; in equal-depth matches at search depth $S \in \{3, 4, 5\}$ with a fixed branching factor, it remains competitive with the configured Stockfish baseline used in our evaluation. Game-level visualizations further show that trajectories form coherent paths through the learned space, indicating that the same representation used for action selection also provides a qualitative account of how positions move between winning, losing, and neutral regions.

The central contribution is therefore not a claim of superiority over full-strength chess engines, but a distinct planning interface. Unlike latent-planning systems that couple learned dynamics with policy or value heads (Hafner et al., 2019; 2020; Schrittwieser et al., 2020), SOLIS performs action selection by applying a fixed geometric scoring rule to embeddings of legal successor states. The advantage vector functions as an evaluation rule induced by supervised representation learning, so the relevant question is whether this rule can support shallow search without a separately trained scalar value model. Chess provides a demanding perfect-information testbed for that question, while the formulation is most naturally relevant to other domains in which informative state evaluations are available or can be learned.

2 BACKGROUND

Planning in very large action spaces is difficult because the set of candidate action sequences expands exponentially with horizon length. For SOLIS, the key design question is how much of this burden must be borne by explicit search and how much can be transferred to a representation whose geometry has already been shaped by evaluation labels. This section situates that design choice with respect to Markov decision processes, model-based reinforcement learning, latent planning, chess-engine design, and contrastive representation learning.

2.1 MARKOV DECISION PROCESSES

Sequential decision-making problems are commonly formalized as Markov Decision Processes (MDPs), represented by the tuple $\{\mathcal{S}, \mathcal{A}, T, R, p(s_0), \gamma\}$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, T the transition model, R the reward function, $p(s_0)$ the initial state distribution, and $\gamma \in [0, 1]$ the discount factor for future rewards (Puterman, 2014). A policy is defined as $\pi : \mathcal{S} \rightarrow p(\mathcal{A})$, mapping each state to a probability distribution over available actions.

Within this formalism, a *model* provides query access to the MDP dynamics: given a state–action pair, it returns or predicts the subsequent state and reward (Moerland et al., 2022). In games such as chess and Go, the rules instantiate this model exactly; in domains where the dynamics are not available, the model must be approximated from data (Gieslmann and Pokorny, 2022).

Planning and reinforcement learning differ primarily in how they exploit, replace, or amortize access to these dynamics (Moerland et al., 2022). Planning methods assume a model and simulate continuations from the current state before selecting an action, yielding a *local* solution that is specific to the decision point and discarded after execution (Russell and Norvig, 1995). Reinforcement-learning methods, by contrast, typically interact with the environment sequentially rather than querying reversible dynamics (Sutton and Barto, 1998); they therefore learn a *global solution*, such as a policy or value function, that generalizes across the state space.

Model-based reinforcement learning combines these regimes by using a model while also learning a reusable global solution. As described by Moerland et al. (2022), this combination has two principal forms: model-based RL with a learned model, such as Dyna (Sutton, 1991), in which both the dynamics and the global solution are estimated; and model-based RL with a known model, such as AlphaZero (Silver et al., 2017), in which the dynamics are supplied and learning concentrates on policy or value estimation.

2.1.1 MODEL-BASED RL WITH A LEARNED MODEL

A substantial body of model-based RL learns both dynamics and a global decision rule. PlaNet (Hafner et al., 2019) learns a latent dynamics model and combines it with learned decision components. Dreamer (Hafner et al., 2020) and its successors DreamerV2 (Hafner et al., 2022) and DreamerV3 (Hafner et al., 2024) extend this paradigm to more complex tasks while retaining global value functions and policies. MuZero (Schrittwieser et al., 2020) also learns latent dynamics, but couples that model to tree search while training a global policy and value function. Dyna (Sutton, 1991) and MBPO (Janner et al., 2021) similarly use learned models to support value or policy learning.

2.1.2 MODEL-BASED RL WITH A KNOWN MODEL

When the environment dynamics are available, algorithmic design can focus on how search and learning should be coupled. AlphaZero (Silver et al., 2017) uses exact game rules as its model and combines Monte Carlo tree search (MCTS) with learned policy and value heads. SAVE (Hamrick et al., 2020) likewise assumes simulator access and integrates Q-learning with MCTS: a learned Q-function supplies a prior for search, and the Q-estimates produced by search are amortized back into the network.

2.1.3 LATENT PLANNING

Latent planning reduces decision complexity by performing search or action selection in a compressed representation space rather than directly over raw states. PlaNet (Hafner et al., 2019), Dreamer (Hafner et al., 2020), and MuZero (Schrittwieser et al., 2020) pair latent dynamics models with value or policy heads, enabling simulated lookahead in the learned space. Expansive Latent Space Trees (ELAST) (Gieselmann and Pokorny, 2022) frame latent planning as exploratory tree search by combining contrastive representations with sampling-based expansion for long-horizon control. Trajectory Autoencoding Planner (TAP) (Jiang et al., 2023) similarly learns a compact latent action space and plans over discrete action codes.

SOLIS occupies a distinct point in this design space. It does not train a policy head, scalar value head, or dynamics model; instead, it distills evaluation targets into the geometry of the representation itself. Inference then selects moves whose successor embeddings advance through regions aligned with favorable outcomes, using only the learned embedding space, a fixed advantage direction, and the legal move generator.

2.2 CHESS ENGINES

Chess has long served as a benchmark for search-intensive artificial intelligence. IBM’s Deep Blue demonstrated the power of large-scale chess computation in 1997 by defeating the reigning world champion, Garry Kasparov, in a head-to-head match. Early versions of Stockfish (Romstad et al., 2008) followed the same broad template: handcrafted evaluation functions combined with deep alpha-beta tree search. This methodology achieved superhuman performance, although its behavior depended substantially on the chess knowledge encoded in the static evaluation function.

The limitations of static evaluation motivated systems that learn evaluation with less direct human guidance. AlphaZero (Silver et al., 2017) and the open-source Leela Chess Zero (Pascutto and Linscott, 2018) demonstrated that engines can acquire strong positional evaluation through self-play reinforcement learning, using no prior knowledge beyond the rules of chess. Modern Stockfish versions similarly incorporate an efficiently updatable neural network evaluation module (NNUE) (Nasu, 2018), improving position evaluation while preserving traditional alpha-beta search.

Although these advances have produced engines far stronger than human players, the underlying search paradigm remains recognizably continuous with principles introduced by v. Neumann (1928) and Shannon (1950) and later engineered at scale in Deep Blue. Strong engines still derive much of their power from deep alpha-beta search over large numbers of states (Lai, 2015). SOLIS asks whether supervised contrastive learning can shift part of that burden from broad explicit expansion to traversal through a high-dimensional representation whose geometry is already aligned with evaluation.

2.3 CONTRASTIVE LEARNING

Contrastive representation learning trains an encoder to place semantically similar inputs near one another and dissimilar inputs farther apart in latent space (Chen et al., 2020). Given a batch I containing an anchor input x_i , an encoder $f(\cdot)$ maps the anchor to a normalized embedding $z_i = f(x_i)$. Positive pairs (z_i, z_j) denote inputs treated as similar under the training criterion, whereas the remaining in-batch samples serve as negatives.

A common contrastive objective is InfoNCE (van den Oord et al., 2019), defined as:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

where $A(i) = I \setminus \{i\}$ is the set of all non-anchor samples in the batch, $\text{sim}(\cdot, \cdot)$ denotes the similarity function, and τ is a scalar temperature parameter. In self-supervised contrastive learning, the positive sample z_j is typically an augmentation of the anchor z_i , so explicit labels are not required (Chen et al., 2020; Khosla et al., 2021).

When labels are available, supervised contrastive learning (SupCon) (Khosla et al., 2021) extends InfoNCE by admitting multiple positive examples for each anchor. The SupCon objective is defined as:

$$\mathcal{L}_{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)}, \quad (2)$$

where $P(i)$ denotes the set of positive examples associated with anchor i , and $A(i)$ is defined as above. Increasing the number of informative positive and negative pairs has been shown to improve representation quality (Khosla et al., 2021; Chowdhury et al., 2024); SOLIS uses this mechanism to turn graded chess evaluations into neighborhoods in latent space.

3 METHODS

We now specify the components of SOLIS: the FEN-based input representation, the transformer encoder, the supervised contrastive training procedure, and the embedding-guided inference rule used for action selection.

3.1 INPUT REPRESENTATION AND TOKENIZATION

Each chess position is represented by its Forsyth-Edwards Notation (FEN) ASCII string, which compactly records piece placement, side to move, castling rights, en passant target, half-move clock, and full-move counter. Following the tokenization scheme of Ruoss et al. (2024), we expand run-length encodings and convert each FEN into a fixed-length sequence of 77 tokens. As noted by Ruoss et al. (2024), this representation omits historical information that can affect future legal or strategic outcomes in special cases, including threefold repetition and the fifty-move rule; consequently, the induced decision problem is not fully Markovian. We return to this limitation in Section 5.

3.2 ENCODER ARCHITECTURE

The SOLIS encoder is a multi-layer transformer (Vaswani et al., 2023) adapted to fixed-length tokenized chess positions. We train two configurations, SOLIS Base and SOLIS Mini, whose architectural parameters are summarized in Table 1.

Table 1: Transformer configurations used for the SOLIS Mini and SOLIS Base encoders.

Model	Layers	Embedding dim. D	Heads	MLP Size	Params
SOLIS Mini	6	128	8	256	0.8M
SOLIS Base	6	1024	16	1024	41M

Each tokenized input is mapped to D -dimensional vectors by a learned token embedding matrix. A classification token (CLS) (Devlin et al., 2019; Dosovitskiy et al., 2021) is prepended to the sequence so that the encoder can aggregate position-level information into a single representation. Because all input sequences have fixed length, we add learned positional encodings to the token embeddings (Ruoss et al., 2024). The sequence is then processed by stacked transformer encoder layers with GELU (Hendrycks and Gimpel, 2023) activations and dropout (Srivastava et al., 2014); the final hidden state of the CLS token is extracted, linearly projected, and ℓ_2 normalized to produce the embedding used for contrastive training and inference.

3.3 SUPERVISED CONTRASTIVE TRAINING

We train the encoder on 5 million randomly sampled positions from ChessBench (Ruoss et al., 2024). Each position is annotated with a Stockfish-evaluated win probability for the player to move, and we normalize all evaluations to White’s perspective so that values near 1.0 indicate a decisive White advantage and values near 0.0 indicate a decisive Black advantage.

Positive pairs are defined by evaluation proximity. Let δ denote the margin used to identify positions with similar win probabilities. For each anchor, we precompute the set of positions whose win probabilities differ by less than δ and randomly sample five positives during training. We also construct the batch-level mask by comparing all in-batch win probabilities and marking a pair as positive when its evaluation difference is below δ . This margin-based criterion pulls together positions with nearly equal evaluations while separating positions with larger outcome gaps, thereby inducing an embedding geometry ordered from losing to winning regions.

We optimize the supervised contrastive objective (SupCon) (Khosla et al., 2021) using these masks to specify positive and negative pairs. Pairwise scores are computed with cosine similarity, and the temperature is fixed at $\tau = 0.07$ (Girdhar et al., 2023; Radford et al., 2021) to control distributional sharpness. Optimization uses stochastic gradient descent with momentum 0.9 (Dosovitskiy et al., 2021; Chen et al., 2020; He et al., 2015), batch size 128, for 400,000 steps. Training is performed with data parallelism on six NVIDIA L40S GPUs (48 GB each) (Paszke et al., 2019). Complete hyperparameters are reported in Appendix A.2.

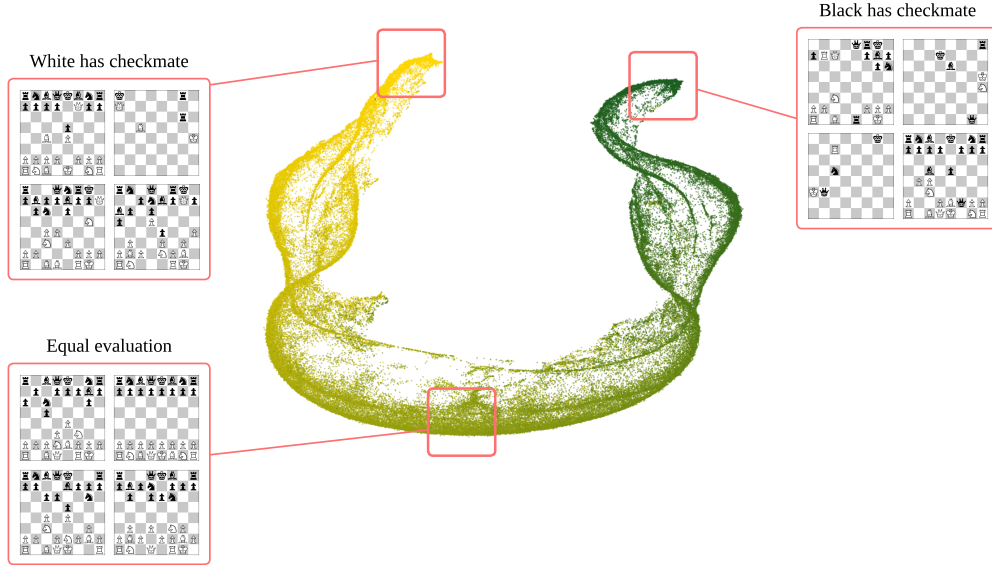


Figure 1: UMAP projection of the SOLIS Base embedding space, with positions colored by Stockfish-evaluated win probability from White’s perspective (gold = White favored, green = Black favored).

Figure 1 visualizes the learned SOLIS Base representation using UMAP (McInnes et al., 2020) for two-dimensional projection. Each point is an encoded position from a test dataset, colored by Stockfish-evaluated win probability from White’s perspective. The resulting structure separates strongly winning White positions, shown in gold (*White has checkmate*), from strongly winning Black positions, shown in dark green (*Black has checkmate*), illustrating that the contrastive objective organizes the latent space according to evaluation.

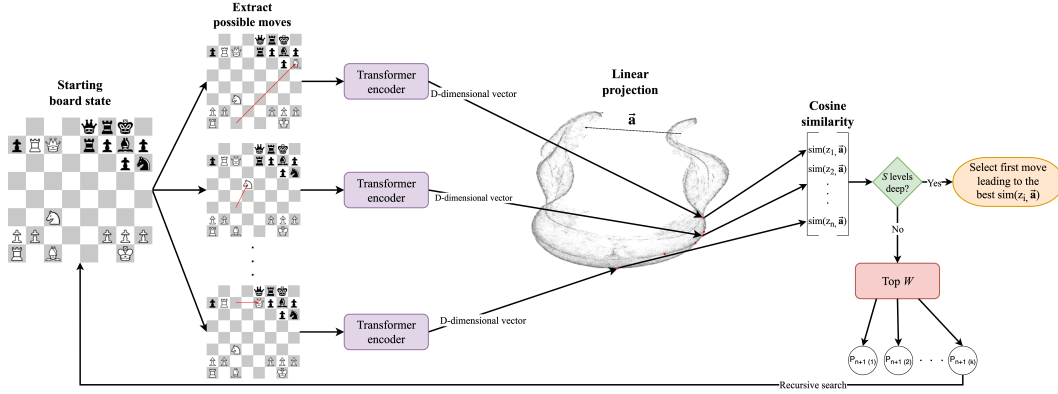


Figure 2: System overview of SOLIS. Legal candidate moves are converted to child positions, embedded by the transformer encoder, and ranked by alignment with the global advantage vector. For simplicity, the diagram illustrates the unanchored score $\cos(z', \vec{a})$.

3.4 EMBEDDING-GUIDED SEARCH

At inference, SOLIS plans directly in the learned embedding space by ranking legal continuations with respect to a single global advantage direction. We estimate this direction from a held-out dataset by computing the mean embeddings at the two evaluation extremes,

$$\mu_{\text{White}} = \mathbb{E}[z \mid p=1.0], \quad \mu_{\text{Black}} = \mathbb{E}[z \mid p=0.0],$$

and define the normalized advantage vector

$$\vec{a} = \frac{\mu_{\text{White}} - \mu_{\text{Black}}}{\|\mu_{\text{White}} - \mu_{\text{Black}}\|}.$$

We evaluate two scoring mechanisms that convert this direction into scalar continuation scores.

3.4.1 UNANCHORED (GLOBAL-DIRECTIONAL)

The unanchored mechanism scores each child embedding by its direct alignment with the global advantage direction. For a child embedding z' , the score is

$$\text{score}_{\text{dir}}(z') = \langle z', \vec{a} \rangle,$$

so the search favors moves whose embeddings lie farthest along the advantage direction, independent of any explicit origin. Figure 2 depicts this scoring mechanism for clarity.

3.4.2 ANCHORED (RELATIVE PROJECTION)

The anchored mechanism first shifts each child embedding by the Black mean and then projects onto the advantage direction:

$$\text{score}_{\text{anch}}(z') = \langle z' - \mu_{\text{Black}}, \vec{a} \rangle,$$

The resulting scalar is proportional to signed distance from μ_{Black} along $\vec{a}$. This formulation removes dependence on an arbitrary coordinate origin and, in our experiments, provides a more stable basis for comparing continuations.

Both scoring mechanisms use the same search procedure (Figure 2). Starting from the “Starting board state”, SOLIS “Extract possible moves”, embeds each child position in parallel with the “Transformer encoder”, and computes either $\text{score}_{\text{dir}}$ or $\text{score}_{\text{anch}}$ in the “Cosine similarity” block. The search retains the “Top- W ” children at each layer. If “ S levels deep?” has been reached, the procedure terminates and “Select first move. . .” is returned; otherwise, SOLIS continues “Recursive search” with min-max preference, where White maximizes the score and Black minimizes it. We use Zobrist hashing (Zobrist, 1990) to maintain a transposition table and avoid redundant position evaluations.

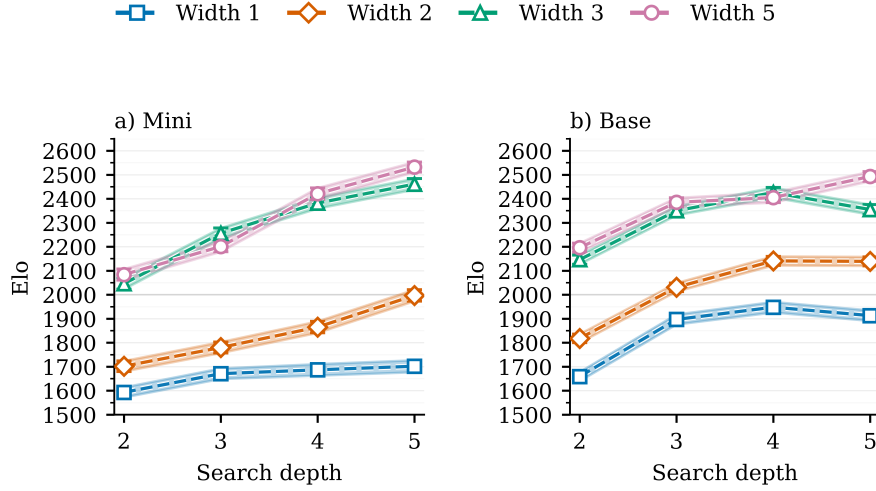


Figure 3: Estimated Elo for SOLIS Mini and SOLIS Base across search widths and depths using the *unanchored* scoring rule. Shaded bands denote 95% confidence intervals.

4 RESULTS

4.1 RATING CALCULATION

We estimate playing strength by evaluating each SOLIS configuration against Stockfish under a fixed per-move time limit of 50 ms (Ruoss et al., 2024); full Stockfish settings are reported in Appendix A.3. The evaluation ablates both search depth and search width. For every depth/width configuration, the corresponding SOLIS model plays 600+ games with alternating colors against Stockfish instances configured at different Elo caps. We evaluate both unanchored and anchored scoring under this protocol. The Stockfish caps are chosen so that each SOLIS configuration has at least one matchup with a positive win rate and one with a negative win rate, enabling rating estimation from both sides of its apparent strength. Elo ratings and 95% CIs are estimated with Bayesian logistic regression using BayesElo (Coulom, 2008) and the default confidence parameter of 0.5. Figure 3 reports unanchored ratings, Figure 4 reports anchored ratings, and Appendix A.6 provides the Stockfish caps and match outcomes required for Elo reproducibility.

Unanchored ratings Under unanchored planning, Elo increases systematically as search becomes deeper. Depth produces the largest gains, whereas width yields a pronounced improvement when moving from $W \in \{1, 2\}$ to $W=3$ and little additional benefit at $W=5$. This plateau indicates that, for this scoring rule, the most valuable continuations are usually already contained among the top three ranked children. At shallow depths, SOLIS Base leads Mini by roughly 100–250 Elo; at depths 4 and 5, Mini becomes competitive with Base and exceeds it for $W \in \{3, 5\}$ under unanchored scoring.

Anchored ratings Anchored scoring produces smooth Elo gains with increasing depth and width, while the Base–Mini gap remains approximately 100–250 Elo. Both models exceed 2500 Elo at depth 5. Unlike the unanchored setting, however, SOLIS Base remains consistently stronger than Mini across the evaluated configurations. Anchoring also improves Base by a steady 10–30 Elo, supporting the interpretation that projection relative to the Black mean yields a more reliable comparison of candidate continuations.

4.2 EQUAL-DEPTH SIMULATIONS

We next compare SOLIS and Stockfish directly at fixed search depths. Guided by the Elo scaling results in Section 4.1, we set the SOLIS branching factor to $W=3$, since wider searches provide only marginal gains and the strongest continuations are generally among the top three candidates.

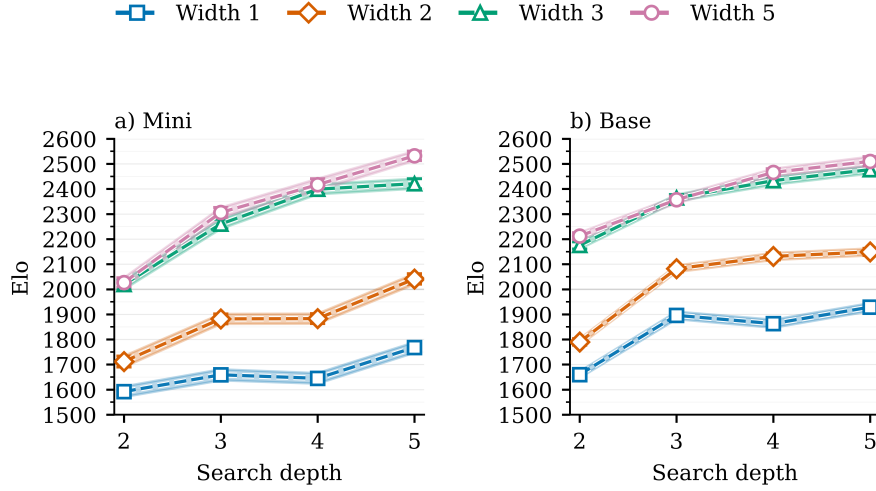


Figure 4: Estimated Elo for SOLIS Mini and SOLIS Base across search widths and depths using the *anchored* projection rule. Shaded bands denote 95% confidence intervals.

Stockfish uses its default expansion strategy and is not constrained to a fixed branching factor. For each depth $S \in \{3, 4, 5\}$, we simulate 200 games with alternating colors. Table 2 reports match points for both unanchored and anchored SOLIS scoring.

Simulation results With unanchored scoring, SOLIS Base scores higher than Stockfish at every tested depth, while SOLIS Mini is competitive at depths 3 and 5 but trails at depth 4. With anchored scoring, Base narrowly trails at depth 3 and then outperforms Stockfish at depths 4 and 5; Mini wins the depth-3 and depth-5 matches but again falls behind at depth 4. These results show that the embedding-guided search remains competitive in equal-depth comparisons even though its branching factor is explicitly capped.

Table 2: Match points for SOLIS vs. Stockfish in 200-game equal-depth matches. The middle columns report unanchored direct-similarity scoring with the global advantage vector, and the right-most columns report anchored projection scoring. Higher scores are better; draws count as 0.5 points.

Config	Depth	Direct Similarity		Anchored Projection	
		SOLIS score	Stockfish score	SOLIS score	Stockfish score
Base	3	106	94	98	102
	4	109	91	108	92
	5	107.5	92.5	110	90
Mini	3	108	92	105	95
	4	85	115	91	109
	5	101	99	113	87

Nodes searched With fixed branching factor $W=3$, the maximum number of SOLIS nodes searched at depth S is $N(S) = \sum_{i=0}^S 3^i$, which gives $N(3) = 40$, $N(4) = 121$, and $N(5) = 364$. Stockfish, by contrast, explores a variable number of nodes per move according to its pruning and search heuristics. In our equal-depth runs, Stockfish searched an average of 137 nodes at depth 3 (maximum 1625), 214 nodes at depth 4 (maximum 4463), and 338 nodes at depth 5 (maximum 10611). SOLIS is therefore competitive with this Stockfish configuration while evaluating substantially fewer positions, indicating that **a substantial portion of the strength normally obtained through broad expansion can be recovered through a higher-quality evaluation-aligned representation.**

4.3 INTERPRETABLE GAME TRAJECTORIES

We visualize sequences of latent positions from real games to assess the interpretability of the learned geometry. Figure 5 presents three representative trajectories: a White win, a draw that remains balanced, and a Black win. Each trajectory is plotted on the same two-dimensional projection of a sample dataset; positions are embedded independently and connected with arrows to show temporal progression. Decisive games tend to move smoothly toward the region associated with the eventual winner, whereas closely contested games remain near the central neutral region. These trajectories complement Figure 1 by showing that the static evaluation structure of the embedding space also supports coherent game-level movement.

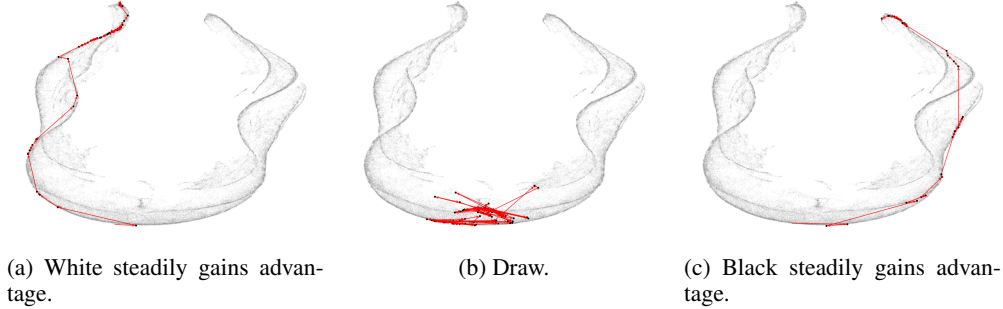


Figure 5: Latent trajectories for three representative games embedded in the shared SOLIS representation space. Red arrows indicate the chronological progression of positions.

5 DISCUSSION

5.1 LIMITATIONS

Blindness to threefold repetition SOLIS makes decisions from the current FEN representation by measuring alignment with the advantage vector, which keeps inference inexpensive but excludes relevant move history. The most important consequence is blindness to threefold repetition: if the same position, including side to move, castling rights, and en-passant rights, occurs three times at any point in the game, either player may claim a draw. An opponent can therefore engineer repeated positions, such as perpetual checks, to neutralize advantages that SOLIS would otherwise be positioned to convert. Game *C* in Appendix A.4 illustrates this failure mode: SOLIS reaches a winning position against Stockfish but permits a premature draw by repetition.

Indecisiveness in winning positions Because SOLIS is trained on win probabilities, it does not distinguish forcing lines that share the same terminal evaluation but differ in length. For example, mate-in-1 and mate-in-20 both have win probability 1.0, even though the shorter continuation is strategically preferable. Choosing a longer winning line can give the opponent additional opportunities to complicate the position, occasionally producing unnecessary draws or losses. Ruoss et al. (2024) observed a related issue and avoided it by consulting Stockfish as an external oracle. SOLIS instead follows its own representation-guided solution, leaving shorter-forced-win preference as an important unresolved limitation. Game *I* in Appendix A.5 shows a case in which SOLIS has many winning continuations over several moves but selects suboptimal moves that lead to a draw.

5.2 CONCLUSION

This paper demonstrates that planning can be performed directly in an evaluation-aligned latent space without training a policy head, value head, or learned dynamics model. SOLIS learns a supervised-contrastive embedding in which proximity reflects outcome similarity and a single global *advantage vector* connects losing and winning regions. Action selection then becomes a vector-space operation: legal continuations are ranked by alignment with this direction and expanded through a deliberately small search.

In chess, SOLIS uses a fixed branching factor $W=3$ and shallow depths $S \in \{3, 4, 5\}$. Under these conditions, it matches or exceeds time-capped Stockfish in several equal-depth matchups while searching far fewer nodes per move (40–364 for SOLIS, compared with hundreds to thousands for Stockfish at the same depths). Elo increases with depth; with anchored scoring, both Mini and Base exceed 2500 Elo at depth five, and anchoring provides a consistent 10–30 Elo improvement for Base. Because additional width beyond three candidates contributes little, the empirical pattern supports the paper’s central claim: much of the search burden can be transferred from broad expansion to the quality of the learned evaluation-aligned space. The embedding also yields interpretable game paths, with decisive games moving toward winning regions and draws remaining near neutral.

Taken together, the results establish evaluation-aligned latent planning as a simple, compute-efficient alternative to approaches that depend on rollout-heavy search, explicit dynamics models, or separately trained policy/value heads. The formulation is most directly applicable to perfect-information settings, and potentially to large-action-space control problems, when informative state evaluations are available or learnable and low-branching search is desirable. Future work should incorporate rule-aware historical features, prefer shorter forced wins, combine SOLIS with policy priors or MCTS, study calibration and uncertainty, and evaluate transfer beyond chess.

REPRODUCIBILITY STATEMENT

Appendix A.2 reports the complete training setup, including hyperparameters and implementation details. Section 3 specifies the input representation, encoder architecture, supervised contrastive objective, and inference procedure, with model configurations summarized in Table 1. Section 4 and Appendix A.3 describe the evaluation protocol, Stockfish settings, and rating calculation procedure. Appendix A.6 provides the detailed match outcomes used to verify the Elo estimates.

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A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models were used only for minor manuscript-support tasks, including formatting and figure/table placement. They were not used to design experiments, analyze results, or generate technical content. All methodological and experimental contributions were developed and verified by the authors.

A.2 TRAINING SETUP

Both SOLIS configurations (Mini and Base) are trained with supervised contrastive learning on 5M ChessBench positions. Training uses cosine similarity over ℓ_2 -normalized embeddings and SGD with momentum on 6 48 GB L40S GPUs for 400k steps with batch size 128.

Among these settings, τ and δ are the most consequential. The temperature $\tau = 0.07$ follows prior contrastive work (Radford et al., 2021; Girdhar et al., 2023), where smaller values sharpen similarity scores and larger ones oversmooth gradients. We trained with $\tau \in \{0.05, 0.07, 0.1\}$ and found 0.07 converged most reliably across model sizes. The margin $\delta = 0.05$ balances collapsing too many positions into one neighborhood (large δ) against fragmenting the space (small δ), and also reflects Stockfish evaluation noise, since sub-0.05 differences are rarely meaningful in practice.

Table 3: Hyperparameters used for supervised-contrastive training of SOLIS Mini and SOLIS Base.

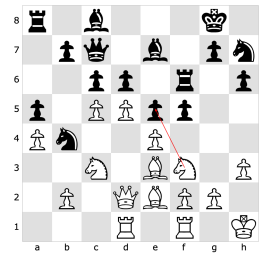
Attribute	Value
Loss	SupCon
τ	0.07
δ (positive margin)	0.05
Positives per anchor	5
Optimizer	SGD
Momentum	0.9
Learning rate	0.05
Batch size	128
Steps	400k
Dropout	0.10
Activation	GELU
Similarity	cosine
Embedding norm	ℓ_2
Token length	77

A.3 STOCKFISH SETUP

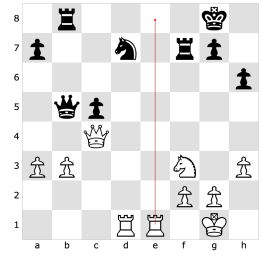
All experiments use the official Linux release of Stockfish 16 (Romstad et al., 2008) with a 32 MB hash size and default threading. Rating caps are imposed through the `UCI_LimitStrength` and `UCI_Elo` settings, and fixed-depth evaluations use the `depth` parameter. To measure the number of nodes searched for each move, we set the engine information level to `INFO_ALL` and extract the `nodes` attribute.

A.4 REPRESENTATIVE GAMES: STOCKFISH (CAP) VS. SOLIS (DEPTH, WIDTH)

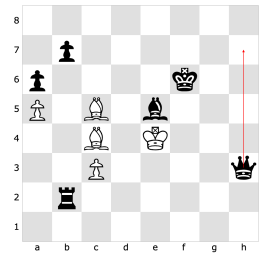
We report representative games between SOLIS and rating-capped Stockfish, with SOLIS constrained to search depth S and width W while Stockfish uses unbounded search under the specified cap. Games *A*, *B*, and *E* illustrate tactical conversion and longer-horizon compensation identified by the latent search procedure. Games *C* and *D* document characteristic failure modes that remain consistent with the limitations discussed above. In each table, the left column gives the game in Portable Game Notation (PGN), and the right column shows a critical position from that game.

1. d4 Nf6 2. c4 d6 3. Nc3 e5 4. Nf3 Qe7 5. e4 c6 6. d5 Qc7 7. Be2 h6 8. a4 a5 9. O-O Be7 10. h3 Na6 11. Be3 Nb4 12. Kh1 Nh7 13. Qd2 O-O 14. Rad1 f5 15. c5 Rf6 16. Nxe5 f4 17. Bxf4 Rxf4 18. Qxf4 dxe5 19. d6 exf4 20. Bc4+ Kf8 21. dxc7 Ke8 22. e5 Ng5 23. Rd6 Bxd6 24. exd6 Bd7 25. Re1+ Kf8 26. Re7 Bf5 27. h4 Nd5 28. Nxd5 cxd5 29. Bxd5 Be6 30. Bxe6 f3 31. hxg5 b6 32. Bh3 hxg5 33. Re5 Rc8 34. Rf5+ Kg8 35. Rf7 g4 36. cxb6 g6 37. d7 fxg2+ 38. Kh2 Rf8 39. Bxg2 Kxf7 40. Bc6 Ke7 41. Kg3 Rf3+ 42. Kh2 Rd3 43. b7 Rh3+ 44. Kg2 Re3 45. d8=Q+ Kf7 46. b8=Q Rg3+ 47. Kh2 Rh3+ 48. Kg1 Rh7 49. Qe8+ Kf6 50. Qbd8+ Kf5 51. Be4+ Kf4 52. Qf6# 1-0	
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Game A — SOLIS ($S=5$, $W=3$) vs. SF 2350, SOLIS–White. Result: 1–0. SOLIS sacrifices a knight to obtain a concrete middlegame advantage and subsequently converts it.

1. d4 d5 2. e3 c6 3. Bd3 h6 4. h3 Nf6 5. Nf3 e6 6. O-O b6 7. c4 Ba6 8. b3 Nbd7 9. Bb2 Bb4 10. a3 Be7 11. Nbd2 O-O 12. Re1 c5 13. Qe2 Qc7 14. e4 Nxe4 15. Nxe4 dxe4 16. Bxe4 Rab8 17. d5 Bf6 18. dxe6 Bxb2 19. exf7+ Rxf7 20. Qxb2 Bb7 21. Bxb7 Qxb7 22. Rad1 b5 23. Qe2 bxc4 24. Qxc4 Qb5 25. Re8+ Nf8 26. Rxb8 Qxb8 27. Rd5 Qf4 28. Qxf4 Rxf4 29. Rxc5 Ne6 30. Rc6 Nd4 31. Nxd4 a5 32. Rc4 Rf8 33. Nc6 Kh8 34. Nxa5 Ra8 35. b4 Rd8 36. g3 Rd3 37. a4 g5 38. b5 Rd2 39. b6 Rd5 40. b7 Rxa5 41. Rc8+ Kh7 42. b8=Q Rd5 43. Rc7+ Kg6 44. Qg8+ Kf5 45. Qxd5+ Kg6 46. Qf7# 1-0	
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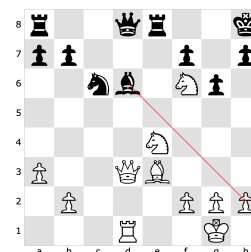
Game B — SOLIS ($S=5$, $W=3$) vs. SF 2300, SOLIS–White. Result: 1–0. SOLIS sacrifices a rook to win Stockfish’s queen and converts the resulting advantage.

1. e3 Nf6 2. Nc3 d5 3. Nf3 c5 4. d4 e6 5. g3 a6 6. a4 Nc6 7. Ne2 Bd6 8. Bg2 O-O 9. h3 Re8 10. c3 Qc7 11. Nd2 e5 12. Nb3 c4 13. dxe5 Nxe5 14. Nbd4 Nd3+ 15. Kf1 Nc5 16. Bf3 Bd7 17. Kg2 Be5 18. a5 Rad8 19. Qc2 g6 20. Bd2 h5 21. Be1 h4 22. gxh4 Kg7 23. Ng3 Rh8 24. h5 Rdg8 25. hxg6 fxg6 26. Be2 Qd6 27. Qd1 Qe7 28. f3 Bd6 29. Nf1 Rh5 30. h4 Rgh8 31. Bf2 R5h7 32. Qe1 Bb8 33. Bg3 Ba7 34. Bf2 Nh5 35. Ng3 Bb8 36. f4 Nf6 37. h5 Rh6 38. hxg6 Rxh1 39. Qxh1 Qe8 40. Ndf5+ Bxf5 41. Nxf5+ Kxg6 42. Ne7+ Kf7 43. e4 Rxh1 44. Rxh1 Qxe7 45. Bxc5 Qxe4+ 46. Bf3 Qg6+ 47. Kf1 Bxf4 48. b4 cxb3 49. Rh8 Ke6 50. Rf8 b2 51. Rxf6+ Kxf6 52. Bxd5 b1=R+ 53. Ke2 Rb2+ 54. Kf3 Be5 55. Bc4 Qh5+ 56. Ke3 Qh3+ 57. Ke4 Qh7+ 58. Kf3 Qh3+ 59. Ke4 Qh7+ 60. Kf3 Qh3+ 61. Ke4 Qh7+ 62. Kf3 Qh3+ 63. Ke4 Qh7+ 64. Kf3 Qh3+ 65. Ke4 1/2-1/2	
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Game C — SF 2200 vs. SOLIS ($S=5$, $W=3$), SOLIS–Black. Result: 1/2–1/2. SOLIS reaches a winning endgame but permits a draw because the current-state representation does not encode repetition history.

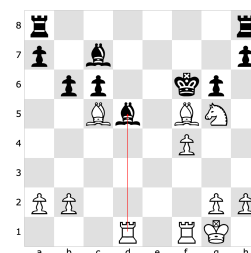
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1. d4 d5 2. c4 dxc4 3. e3 e5 4. Bxc4 exd4 5. exd4
Nf6 6. Nc3 Bd6 7. Qe2+ Qe7 8. Be3 O-O 9. Nf3 c6 10.
O-O Be6 11. Rfd1 Re8 12. d5 cxd5 13. Bxd5 Nc6 14.
a3 Nxd5 15. Nxd5 Qd8 16. Qd3 Bg4 17. Ng5 g6 18. Ne4
Bxd1 19. Ndf6+ Kh8 20. Rxd1 Bxh2+ 21. Kxh2 Qxd3 22.
Rxd3 Red8 23. Nd7 Kg7 24. f3 h6 25. Rd6 f5 26. Nec5
Re8 27. Bd2 Re2 28. Kh1 Kf7 29. b3 Ke7 30. Nxb7 Rc8
31. Ndc5 Kf7 32. Kg1 Kg8 33. Rxg6+ Kf7 34. Rd6 Ne7
35. Rd7 Rg8 36. Nd6+ Kg6 37. Nc4 Kh5 38. Ne3 f4 39.
Ne6 fxe3 40. Nf4+ Kg5 41. Nxe2 exd2 42. Kf2 Kf6 43.
Nc3 Ke6 44. Rd4 Rc8 45. Ne4 d1=N+ 46. Rxd1 Rc2+ 47.
Kg1 Nf5 48. Rd2 Rxd2 49. Nxd2 Ke5 50. Kh2 Kf4 51.
b4 Ke3 52. Ne4 Nd4 53. a4 Nc2 54. b5 Nd4 55. Nd6
Ne6 56. a5 Kd3 57. g4 Kd4 58. b6 axb6 59. axb6 Nc5
60. Kg3 Kd5 61. b7 Na6 62. Nf7 Kc6 63. Nxb6 Kxb7
64. f4 Kc7 65. Kh4 Kd7 66. Kh5 Nc5 67. f5 Kd6 68.
f6 Ne6 69. Nf7+ Kc6 70. Kh6 Kd5 71. g5 Nc5 72. Kg6
Ne6 73. Kh5 Kd4 74. g6 Kc4 75. Ne5+ Kd5 76. Ng4 Ke4
77. Kh6 Kf5 78. g7 Nxg7 79. Kxg7 Kxg4 80. Kf7 Kf5
81. Ke7 Kg5 82. f7 Kf4 83. Kf6 Kf3 84. Ke5 Kg3 85.
Ke4 Kf2 86. Kd3 Kg1 87. Ke3 Kh2 88. f8=Q Kg3 89.
Qg8+ Kh3 90. Kf4 Kh2 91. Kf3 Kh3 92. Qg3# 1-0



Game D — SF 2500 vs. SOLIS ($S=5, W=3$), SOLIS–Black. Result: 1-0 SOLIS selects a locally attractive sacrifice that produces a persistent material disadvantage.

1. d4 d5 2. e3 Nd7 3. c4 e5 4. dxe5 dxc4 5. Nf3
Qe7 6. Bxc4 Nxe5 7. Nxe5 Qxe5 8. Qb3 Qh5 9. e4
Nf6 10. O-O Bd6 11. Qb5+ c6 12. Qxh5 Nxb5 13. Nc3
Be5 14. f4 Bc7 15. Be3 g6 16. Bb3 Be6 17. Rad1 Ng7
18. Bc2 f5 19. Bc5 Kf7 20. exf5 Nxf5 21. Ne4 b6
22. Ng5+ Kf6 23. Bxf5 Bd5 24. Rxd5 cxd5 25. Bd4+
Kxf5 26. h3 Bxf4 27. Bxb8 Kxg5 28. Bg7 Rg8 29. Bd4
Re8 30. Rf3 Re1+ 31. Kf2 Re6 32. g3 Bd2 33. Rf7
Re4 34. Bf6+ Kh6 35. g4 a6 36. Kg3 d4 37. h4 Be1+
38. Kh3 Re3+ 39. Kg2 Bxb4 40. Bxb4 a5 41. Bf6 a4
42. Rd7 Rd3 43. Rd5 b5 44. Rxb5 a3 45. Re5 Rd2+
46. Kf3 axb2 47. g5+ Kh5 48. Rb5 Rd3+ 49. Kf4 h6
50. gxh6+ Kxh6 51. Rxb2 g5+ 52. Kg4 Rd1 53. Bxg5+
Kg6 54. Rb6+ Kg7 55. Kf5 Kf7 56. Rb7+ Ke8 57. a4
Ra1 58. Re7+ Kf8 59. Rh7 Rxa4 60. Bh6+ Kg8 61. Re7
Rc4 62. Kg5 Rc5+ 63. Kg4 Ra5 64. Rg7+ Kh8 65. Re7
d3 66. Rd7 d2 67. Bxd2 Rc5 68. Bf4 Kg8 69. Be3 Rb5
70. Bh6 Re5 71. Kf4 Rc5 72. Ke4 Rb5 73. Kf4 Rb4+
74. Ke5 Rb1 75. Kf6 Rb2 76. Kg6 Rb6+ 77. Kh5 Rb2
78. Bg5 Rb6 79. Bh6 Re6 80. Kg5 Ra6 81. Rg7+ Kh8
82. Re7 Ra5+ 83. Kg6 Ra6+ 84. Kh5 Kg8 85. Rb7 Rc6
86. Rd7 Rc5+ 87. Bg5 Rc2 88. Kg6 Rc6+ 89. Bf6 Re6
90. Rd8+ Re8 91. Rxe8# 1-0

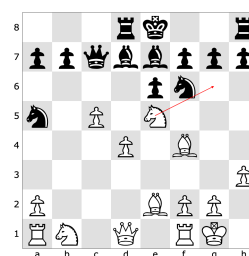


Game E — SOLIS ($S=5, W=3$) vs. Stockfish 2500 , SOLIS–White. Result: 1-0 SOLIS makes an exchange sacrifice for long-term compensation and converts the resulting advantage.

A.5 EQUAL-DEPTH MATCHES: SOLIS D vs. STOCKFISH D

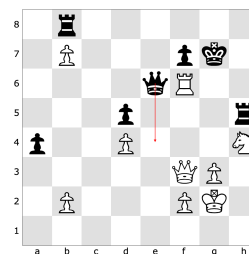
We also include illustrative equal-depth games in which SOLIS and Stockfish both search to depth D . These examples show how the fixed-width latent planning procedure behaves in direct head-to-head play without imposing a rating cap on Stockfish. As above, the left column records the moves in Portable Game Notation (PGN), and the right column displays a critical position.

1. d4 d5 2. e3 c5 3. Nf3 Nf6 4. Be2 e6 5. O-O Nc6
6. b3 cxd4 7. exd4 Be7 8. c3 Qc7 9. h3 Bd7 10. c4
dxc4 11. bxc4 Na5 12. Ne5 Rd8 13. Bf4 Bd6 14. c5
Be7 15. Ng6 Qc8 16. Nxh8 Bc6 17. Nc3 Kf8 18. Rc1
Qa8 19. Bc7 b5 20. Bxa5 Bxg2 21. d5 Bxh3 22. Qd3
Bf5 23. Qg3 b4 24. Rfd1 Bxc5 25. Qc7 Be7 26. Nb5
Nxd5 27. Rxd5 Qxd5 28. Bxb4 Qd7 29. Qxd7 Re8 30.
Rc8 g5 31. Qxe7+ Kg7 32. Qxf7+ Kh6 33. Bf8+ Rxf8 34.
Qxf8# 1-0



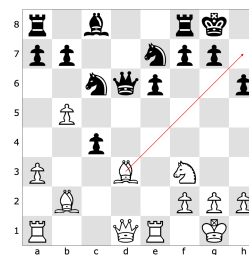
Game F — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 SOLIS sacrifices a knight in the middlegame to secure a concrete tactical advantage.

1. d4 d5 2. e3 c5 3. Nf3 e6 4. Be2 cxd4 5. exd4
Bd6 6. O-O Nf6 7. Bg5 Nbd7 8. Nbd2 h6 9. Bh4 O-O
10. Re1 a5 11. Bf1 Bf4 12. c3 Qb6 13. Qc2 Qc7 14.
a4 b6 15. Bb5 Ba6 16. Bxa6 Rxa6 17. Qd3 Ra7 18. Re2
Qc8 19. Reel Nh5 20. Nf1 Nh6 21. N1d2 Nh5 22. Nf1
Bc7 23. Qd1 Nh6 24. Qd3 Nh5 25. Qd1 Nh6 26. Qd3
Rb7 27. Qb5 Ra7 28. N1d2 Nh5 29. h3 g5 30. Rac1
gxh4 31. c4 Nf4 32. Rc3 Qd8 33. g3 Nxh3+ 34. Kg2
Ng5 35. cxd5 h3+ 36. Kg1 h2+ 37. Kxh2 exd5 38. Rc6
Nf6 39. Nxg5 hxg5 40. Kg2 Re8 41. Rh1 Bd6 42. Nf3
Kg7 43. Nxg5 Rc7 44. Rxd6 Qxd6 45. Qf1 b5 46. axb5
Rcc8 47. Rh4 Rh8 48. Nf3 Ne4 49. Rf4 Rh5 50. Qe2
Qe6 51. Nh4 Rch8 52. b6 Nf6 53. Qf3 a4 54. b7 Rb8
55. Rxf6 Qe4 56. Rxf7+ Kh8 57. Qxe4 dxe4 58. Rc7
Rg5 59. Rc8+ Rg8 60. Rc7 Rg5 61. Rc8+ Rg8 62. Ng6+
Kg7 63. Ne7 Rf8 64. Rc7 Rf7 65. d5 Rbf8 66. Nf5+
Kg6 67. Rxf7 Kxf7 68. Nd6+ Kg6 69. Nc8 Kg5 70. b8=Q
Kg4 71. Qc7 Rf5 72. Nd6 e3 73. Qg7+ Rg5 74. f3+ Kh5
75. Qh7# 1-0



Game G — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 Stockfish builds an advantage but blunders in the late middlegame; SOLIS identifies the opportunity and converts in the endgame.

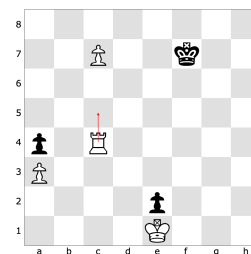
1. d4 c6 2. e4 d5 3. Nc3 Nf6 4. e5 Ng8 5. Bd3 e6
6. Nf3 c5 7. dxc5 Bxc5 8. O-O Nc6 9. a3 h6 10. b4
Bb6 11. Nb5 Bc7 12. Re1 Bb8 13. c4 Nge7 14. Nd6+
Bxd6 15. exd6 Qxd6 16. Bb2 O-O 17. b5 dxc4 18. Bh7+
Kh8 19. Qxd6 f6 20. Be4 c3 21. Bxc3 e5 22. bxc6 Bf5
23. Qxe7 Bxe4 24. Rxe4 bxc6 25. Nxe5 Rfe8 26. Nf7+
Kg8 27. Nxh6+ Kh7 28. Rae1 Rad8 29. Ba5 Rd4 30. g4
Rxe4 31. Rxe4 Rb8 32. Qxa7 Rd8 33. Rf4 Rd3 34. g5
fxg5 35. Rd4 Rxa3 36. Kf1 Rb3 37. Rb4 Rc3 38. h4
gxh4 39. Rd4 h3 40. Qe7 h2 41. Qe4+ Kxh6 42. Rd6+
Kg5 43. Rg6+ Kh5 44. Qg4# 1-0



Game H — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1-0 SOLIS sacrifices a bishop to create a discovered attack on Stockfish's queen and converts the advantage.

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1. d4 d5 2. e3 c6 3. c4 e6 4. Nf3 Nf6 5. Be2 c5
6. O-O Nc6 7. b3 a6 8. Bb2 Be7 9. a3 cxd4 10. exd4
O-O 11. c5 b6 12. b4 bxc5 13. dxc5 Bb7 14. Nbd2 a5
15. b5 Bxc5 16. Rc1 Qb6 17. bxc6 Ng4 18. Rxc5 Ba6
19. Rc2 Nxf2 20. Rxf2 Bxe2 21. Qxe2 a4 22. c7 Rac8
23. Bd4 Qb5 24. Qxb5 Rfe8 25. Qb8 h6 26. Ne5 g5 27.
h4 Rf8 28. Ne4 dxe4 29. Rxf7 Rxb8 30. Rxf8+ Rxf8 31.
hxcg5 Rc8 32. gxh6 e3 33. Bxe3 Kh7 34. Rc4 Kg8 35.
Kh2 Kh7 36. g4 Rg8 37. g5 Ra8 38. Rg4 Kh8 39. Ba7
Rf8 40. Bb8 Rf2+ 41. Kg1 Rc2 42. Rc4 Rb2 43. Ba7
Rb1+ 44. Kg2 Kh7 45. Bf2 Rb2 46. g6+ Kxh6 47. Kh2
Rxf2+ 48. Kg3 Rg2+ 49. Kxg2 Kg7 50. Rc1 Kf6 51. Kh2
Kg7 52. Nd7 e5 53. Nf6 Kxf6 54. Rg1 Kg7 55. Rf1 Kh6
56. Rg1 Kg7 57. Rf1 Kxg6 58. Rc1 Kg7 59. Rc4 Kf7
60. Kg2 Ke7 61. Kf2 Kd6 62. Ke1 Kd5 63. Rc1 Ke4 64.
Ke2 Kd4 65. Rc2 e4 66. Rc1 Kd5 67. Ke1 Ke5 68. Rc4
Kd5 69. Rc2 Kd6 70. Rc1 Ke7 71. Rc4 Kf7 72. Rc5 e3
73. Kf1 Ke7 74. Rc4 Kf7 75. Kg2 e2 76. Kf2 Ke7 77.
Ke1 Kf7 78. Rc5 Kg7 79. Rc2 Kf6 80. Rc4 Kf7 81. Rc5
Kg6 82. Kd2 Kf7 83. Ke1 Ke7 84. Rc4 Kf7 85. Rc5 Kg6
86. Rc4 Kf7 87. Rc5 1/2-1/2



Game I — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1/2-1/2 SOLIS fails to prefer the most direct winning trajectory when many continuations share a winning evaluation.

A.6 SOLIS VS. STOCKFISH RESULTS

To make the Elo estimates reproducible, we report the full match outcomes between SOLIS and Stockfish for SOLIS search depths $S \in \{2, 3, 4, 5\}$. Each table is indexed by Stockfish Elo cap (rows) and SOLIS search width W (columns). Entries give win–loss–draw counts for the corresponding matchup, with total match points in parentheses. Separate tables are provided for the Base and Mini models under both unanchored and anchored scoring.

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 6: Base model vs. Stockfish match outcomes by search depth and Stockfish Elo cap under unanchored planning. Entries report W–D–L counts with total score in parentheses out of 100 games.

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980
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1001
1002
1003
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1007
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1012
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1015
1016
1017
1018
1019
1020
1021
1022
1023
1024
1025

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	32-32-36 (48)	27-45-28 (50)	-	-	1350	-	-	-	-
1400	30-41-29 (50)	33-27-40 (46)	-	-	1400	-	-	-	-
1450	29-44-27 (51)	33-40-27 (53)	-	-	1450	-	-	-	-
1500	31-38-31 (50)	31-39-30 (50)	-	-	1500	29-38-33 (48)	28-43-29 (50)	-	-
1550	38-35-27 (56)	23-50-27 (48)	-	-	1550	41-36-23 (59)	36-37-27 (54)	-	-
1600	34-30-36 (49)	29-37-34 (48)	-	-	1600	37-30-33 (52)	37-30-33 (52)	-	-
1650	24-45-31 (46)	28-36-36 (46)	-	-	1650	32-38-30 (51)	30-40-30 (50)	-	-
1700	-	-	-	-	1700	28-37-35 (46)	21-43-36 (42)	-	-
1750	-	-	-	-	1750	36-33-31 (52)	28-40-32 (48)	-	-
1800	-	-	37-31-32 (52)	37-36-27 (55)	1800	39-24-37 (51)	29-35-36 (46)	-	-
1850	-	-	34-24-42 (46)	34-31-35 (50)	1850	-	-	-	-
1900	-	-	40-28-32 (54)	40-29-31 (54)	1900	-	-	-	-
1950	-	-	39-29-32 (54)	39-29-32 (54)	1950	-	-	-	-
2000	-	-	38-34-28 (55)	42-28-30 (56)	2000	-	-	34-35-31 (52)	38-21-41 (48)
2050	-	-	41-27-32 (54)	40-26-34 (53)	2050	-	-	38-34-28 (55)	34-33-33 (50)
2100	-	-	35-30-35 (50)	36-26-38 (49)	2100	-	-	33-34-33 (50)	34-29-37 (48)
2150	-	-	-	-	2150	-	-	31-35-34 (48)	38-25-37 (50)
2200	-	-	-	-	2200	-	-	38-18-44 (47)	31-26-43 (44)
2250	-	-	-	-	2250	-	-	36-23-41 (48)	45-26-29 (58)
2300	-	-	-	-	2300	-	-	40-22-38 (51)	34-26-40 (47)
2350	-	-	-	-	2350	-	-	-	-
2400	-	-	-	-	2400	-	-	-	-
2500	-	-	-	-	2500	-	-	-	-
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	-	-	-	-	1350	-	-	-	-
1400	-	-	-	-	1400	-	-	-	-
1450	-	-	-	-	1450	-	-	-	-
1500	-	-	-	-	1500	-	-	-	-
1550	-	-	-	-	1550	-	-	-	-
1600	36-42-22 (57)	35-31-34 (50)	-	-	1600	-	-	-	-
1650	29-40-31 (49)	36-33-31 (52)	-	-	1650	-	-	-	-
1700	33-34-33 (50)	36-32-32 (52)	-	-	1700	39-30-31 (54)	34-34-32 (51)	-	-
1750	43-26-31 (56)	28-38-34 (47)	-	-	1750	37-29-34 (52)	32-30-38 (47)	-	-
1800	38-24-38 (50)	30-37-33 (48)	-	-	1800	42-20-38 (52)	33-31-36 (48)	-	-
1850	49-23-28 (60)	29-33-38 (46)	-	-	1850	36-24-40 (48)	43-28-29 (57)	-	-
1900	-	-	-	-	1900	45-26-29 (58)	46-31-23 (62)	-	-
1950	-	-	-	-	1950	40-22-38 (51)	37-35-28 (54)	-	-
2000	-	-	-	-	2000	-	-	-	-
2050	-	-	-	-	2050	-	-	-	-
2100	-	-	-	-	2100	-	-	-	-
2150	-	-	31-31-38 (46)	44-21-35 (54)	2150	-	-	-	-
2200	-	-	35-29-36 (50)	48-27-25 (62)	2200	-	-	27-37-36 (46)	33-38-29 (52)
2250	-	-	37-31-32 (52)	37-28-35 (51)	2250	-	-	30-33-37 (46)	38-31-31 (54)
2300	-	-	27-37-36 (46)	36-34-30 (53)	2300	-	-	38-37-25 (56)	33-39-28 (52)
2350	-	-	22-42-36 (43)	37-35-28 (54)	2350	-	-	32-29-39 (46)	29-36-35 (47)
2400	-	-	36-29-35 (50)	37-26-37 (50)	2400	-	-	30-26-44 (43)	30-32-38 (46)
2500	-	-	-	-	2500	-	-	26-34-40 (43)	43-25-32 (56)

Figure 7: Mini model vs. Stockfish match outcomes by search depth and Stockfish Elo cap under unanchored planning. Entries report W-D-L counts with total score in parentheses out of 100 games.

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1079

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–
Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 8: Base model vs. Stockfish match outcomes by search depth and Stockfish Elo cap under anchored planning. Entries report W–D–L counts with total score in parentheses out of 100 games.

1080
1081
1082
1083
1084
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1097
1098
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1102
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1104
1105
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1110
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1112
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1117
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1120
1121
1122
1123
1124
1125
1126
1127
1128
1129
1130
1131
1132
1133

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	36–32–32 (52)	39–30–31 (54)	–	–	1350	–	–	–	–
1400	23–49–28 (48)	30–32–38 (46)	–	–	1400	–	–	–	–
1450	33–37–30 (52)	24–47–29 (48)	–	–	1450	–	–	–	–
1500	34–38–28 (53)	25–40–35 (45)	–	–	1500	29–40–31 (49)	35–33–32 (52)	–	–
1550	28–38–34 (47)	40–39–21 (60)	–	–	1550	34–35–31 (52)	37–29–34 (52)	–	–
1600	23–47–30 (46)	28–42–30 (49)	–	–	1600	32–36–32 (50)	32–39–29 (52)	–	–
1650	35–29–36 (50)	38–37–25 (56)	–	–	1650	34–30–36 (49)	37–33–30 (54)	–	–
1700	–	–	–	–	1700	36–30–34 (51)	34–35–31 (52)	–	–
1750	–	–	–	–	1750	34–25–41 (46)	34–30–36 (49)	–	–
1800	–	–	35–37–28 (54)	40–28–32 (54)	1800	32–29–39 (46)	32–31–37 (48)	–	–
1850	–	–	32–32–36 (48)	44–29–27 (58)	1850	–	–	–	–
1900	–	–	37–30–33 (52)	45–21–34 (56)	1900	–	–	–	–
1950	–	–	43–30–27 (58)	36–29–35 (50)	1950	–	–	–	–
2000	–	–	35–31–34 (50)	43–27–30 (56)	2000	–	–	46–32–22 (62)	38–27–35 (52)
2050	–	–	37–29–34 (52)	45–27–28 (58)	2050	–	–	37–27–36 (50)	39–26–35 (52)
2100	–	–	36–27–37 (50)	42–34–24 (59)	2100	–	–	29–34–37 (46)	48–17–35 (56)
2150	–	–	–	–	2150	–	–	37–33–30 (54)	39–25–36 (52)
2200	–	–	–	–	2200	–	–	37–31–32 (52)	41–23–36 (52)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	44–20–36 (54)
2300	–	–	–	–	2300	–	–	30–43–27 (52)	38–26–36 (51)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	26–45–29 (48)	31–33–36 (48)	–	–	1600	–	–	–	–
1650	37–39–24 (56)	28–32–40 (44)	–	–	1650	–	–	–	–
1700	23–39–38 (42)	29–41–30 (50)	–	–	1700	37–36–27 (55)	34–29–37 (48)	–	–
1750	36–29–35 (50)	38–30–32 (53)	–	–	1750	42–23–35 (54)	37–33–30 (54)	–	–
1800	35–24–41 (47)	33–36–31 (51)	–	–	1800	36–28–36 (50)	31–35–34 (48)	–	–
1850	41–21–38 (52)	30–28–42 (44)	–	–	1850	38–21–41 (48)	33–32–35 (49)	–	–
1900	–	–	–	–	1900	37–27–36 (50)	37–37–26 (56)	–	–
1950	–	–	–	–	1950	43–18–39 (52)	38–24–38 (50)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	39–23–38 (50)	32–30–38 (47)	2150	–	–	–	–
2200	–	–	33–40–27 (53)	38–25–37 (50)	2200	–	–	30–40–30 (50)	32–34–34 (49)
2250	–	–	36–31–33 (52)	30–27–43 (44)	2250	–	–	29–43–28 (50)	49–26–25 (62)
2300	–	–	32–33–35 (48)	37–28–35 (51)	2300	–	–	31–30–39 (46)	30–35–35 (48)
2350	–	–	35–34–31 (52)	40–25–35 (52)	2350	–	–	26–32–42 (42)	40–34–26 (57)
2400	–	–	28–26–46 (41)	26–38–36 (45)	2400	–	–	39–32–29 (55)	31–32–37 (47)
2500	–	–	–	–	2500	–	–	30–29–41 (44)	28–41–31 (48)

Figure 9: Mini model vs. Stockfish match outcomes by search depth and Stockfish Elo cap under anchored planning. Entries report W–D–L counts with total score in parentheses out of 100 games.